

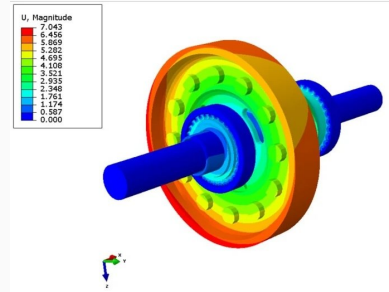
Finite element method

Linear elastodynamics

ME473 Dynamic finite element analysis of structures

Stefano Burzio

2025



Where do we stand?

Week	Module	Lecture topic	Mini-projects
1	Linear elastodynamics	Strong and weak forms	
2		Galerkin and FE methods	
3		Systematisation of the procedure	Groups formation
4		3d elements, numerical integration	Project 1 statement

Summary

- Recap week 3
- Solid finite elements
- Numerical integration
- MATLAB PDE Toolbox examples
- Abaqus and Ansys examples

Recommended readings

- ① Gmür, Dynamique des structures (§3.1, §3.2, and §3.3)

Recap week 3

Finite element method local approach

Localization and assembly

- **Localization:** real and virtual displacements are restricted to each finite element and are expressed in terms of local shape functions and nodal displacements:

$${}^e\mathbf{u}^h(\mathbf{x}, t) = {}^e\mathbf{H}(\mathbf{x}){}^e\mathbf{q}(t) \quad \text{and} \quad {}^e\delta\mathbf{u}^h(\mathbf{x}) = {}^e\mathbf{H}(\mathbf{x}){}^e\delta\mathbf{q}.$$

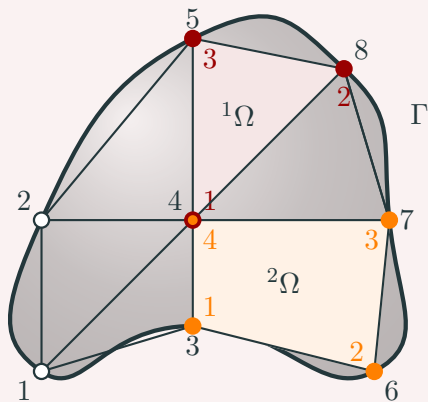
- **Localization matrices** ${}^e\mathbf{L}$: boolean matrices that map local to global node numbering.
- **Elementary quantities:** elementary stiffness matrix ${}^e\mathbf{K}$, elementary mass matrix ${}^e\mathbf{M}$, and elementary force vector ${}^e\mathbf{r}$ are computed for each element.
- **Assembly:** global matrices and vectors are assembled using the localization matrices:

$$\mathbf{K} = \bigcup_{e=1}^m {}^e\mathbf{K} = \sum_{e=1}^m {}^e\mathbf{L}^T {}^e\mathbf{K} {}^e\mathbf{L} \quad \text{and} \quad \mathbf{r} = \bigcup_{e=1}^m {}^e\mathbf{r} = \sum_{e=1}^m {}^e\mathbf{L}^T {}^e\mathbf{r}$$

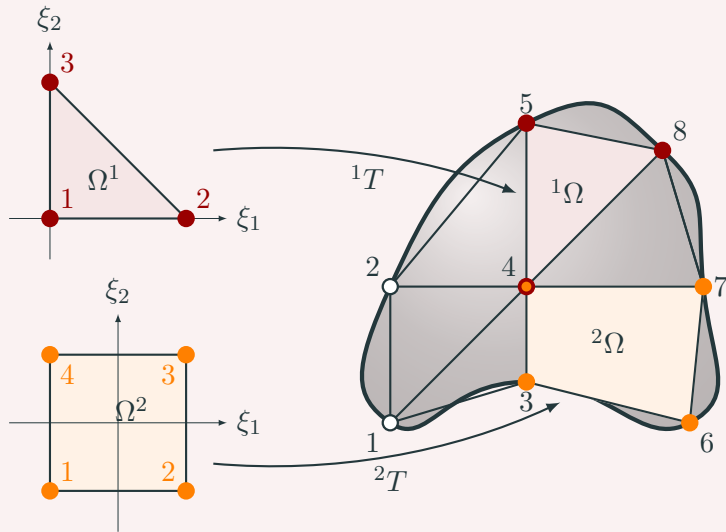
The local finite element point of view (in 2d)

Every finite element ${}^e\Omega$ has:

- Number of nodes ep ,
- Connectivity (${}^ep \times 1$),
- ${}^e\mathbf{q}$ vector of nodal displacements ($2{}^ep \times 1$),
- ${}^e\mathbf{u}$ vector of displacements (2×1),
- eh_i local shape functions ($i = 1, \dots, {}^ep$),
- ${}^e\mathbf{H}$ local shape functions matrix ($2 \times 2{}^ep$),
- ${}^e\mathbf{K}$ local stiffness matrix ($2{}^ep \times 2{}^ep$),
- ${}^e\mathbf{M}$ local mass matrices ($2{}^ep \times 2{}^ep$),
- ${}^e\mathbf{r}$ local loads vector ($2{}^ep \times 1$).



Archetypal to local



Paradigm shift: many with few

- 👉 **Main idea:** instead of working with local shape functions ${}^e h_i$ (one per node!) we will deduce them from a **small set of archetypal (or master) shape functions** ${}^a h_i$ defined on an archetypal element via ${}^e T$.

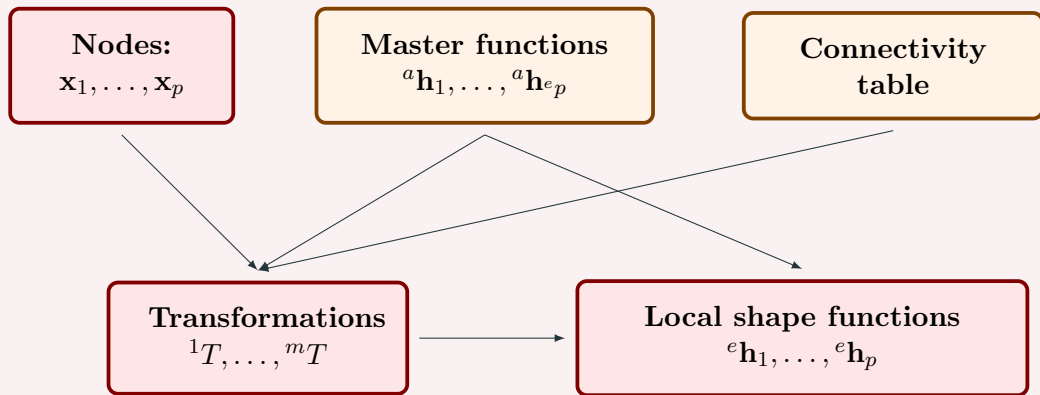


- 👉 **Clever idea:** ${}^e T$ itself is defined in terms of archetypal shape functions and nodes coordinates:

$${}^e T : \mathbf{x}(\boldsymbol{\xi}) = {}^a \mathbf{H}(\boldsymbol{\xi}) {}^e \mathbf{x} = \sum_{i=1}^{e_p} {}^a h_i(\boldsymbol{\xi}) {}^e \mathbf{x}_i$$

- 👉 Archetypal elements follow *established conventions* (node numbering and nodes coordinates) that one needs to adopt.

The clever idea



Concrete example: bilinear triangular element (2d)

Node id local	Local Coord	Node id global	Global coord.
1	(0,0)	4	(x_4, y_4)
2	(1,0)	8	(x_8, y_8)
3	(0,1)	5	(x_5, y_5)

Base (archetypal) functions:

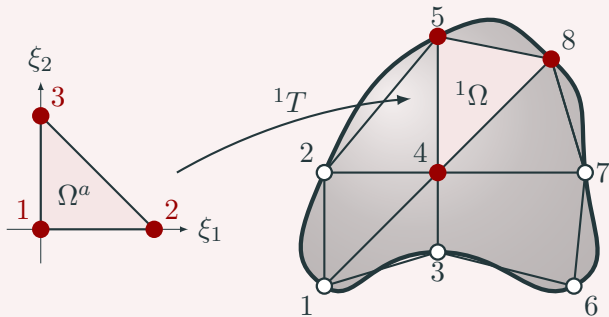
$$^a h_1 = 1 - \xi_1 - \xi_2$$

$$^a h_2 = \xi_1$$

$$^a h_3 = \xi_2$$

Transformation:

$$^1 T : \begin{cases} x(\xi_1, \xi_2) = x_4 h_1 + x_8 h_2 + x_5 h_3 = x_4 + (x_8 - x_4)\xi_1 + (x_5 - x_4)\xi_2 \\ y(\xi_1, \xi_2) = y_4 h_1 + y_8 h_2 + y_5 h_3 = y_4 + (y_8 - y_4)\xi_1 + (y_5 - y_4)\xi_2 \end{cases}$$



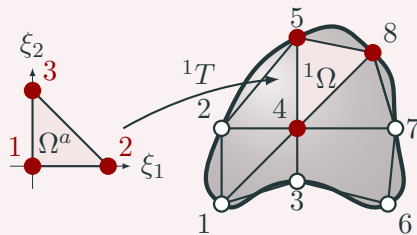
Consequence 1: explicit use of local shape functions is avoided

- The coordinate transformation ${}^eT^{-1}$ maps shape functions on ${}^e\Omega$ to the master element space Ω^a :

$${}^e\mathbf{H}(\mathbf{x}) = {}^a\mathbf{H}({}^eT^{-1}(\mathbf{x}))$$

- An example of a relationship between local and archetypal shape functions:

$${}^1h_8(x, y) = {}^ah_2({}^1T^{-1}(x, y)) = {}^ah_2(\xi_1(x, y), \xi_2(x, y))$$



- 👉 In general, the inverse transformations ${}^eT^{-1} : (\xi_1(x, y), \xi_2(x, y))$ are not easy to compute since the expressions in $\mathbf{x}(\boldsymbol{\xi})$ are nonlinear.
- 👉 Although the local functions eh_j can have intricate expressions, their explicit computation is not required!

Consequence 2: integration over archetypal elements

Thanks to eT , the integrals over ${}^e\Omega$ in the definitions of elementary stiffness and mass matrices and elementary loads vectors can be computed over the regular element Ω^a by performing a change of variables:

$$\begin{aligned}\int_{{}^e\Omega} (\dots) d\Omega &\xrightarrow{\text{replaced by}} \int_{\Omega^a} (\dots) {}^ej d\xi_1 d\xi_2 \\ \frac{\partial}{\partial x_i} &\xrightarrow{\text{replaced by}} \frac{\partial \xi_1}{\partial x_i} \frac{\partial}{\partial \xi_1} + \frac{\partial \xi_2}{\partial x_i} \frac{\partial}{\partial \xi_2}\end{aligned}$$

where ej is the determinant of the Jacobian matrix ${}^e\mathbf{J}$ of the transformation eT .



For each element, it is necessary to compute the transformation, the Jacobian matrix, its determinant, and its inverse.

Concrete example: bilinear triangular element (2d)

■ Transformation:

$${}^1T : \begin{cases} x(\xi_1, \xi_2) = x_4 h_1 + x_8 h_2 + x_5 h_3 = x_4 + (x_8 - x_4)\xi_1 + (x_5 - x_4)\xi_2 \\ y(\xi_1, \xi_2) = y_4 h_1 + y_8 h_2 + y_5 h_3 = y_4 + (y_8 - y_4)\xi_1 + (y_5 - y_4)\xi_2 \end{cases}$$

■ Jacobian matrix:

$${}^1\mathbf{J} = \begin{bmatrix} \partial x / \partial \xi_1 & \partial x / \partial \xi_2 \\ \partial y / \partial \xi_1 & \partial y / \partial \xi_2 \end{bmatrix} = \begin{bmatrix} x_8 - x_4 & x_5 - x_4 \\ y_8 - y_4 & y_5 - y_4 \end{bmatrix}$$

■ Jacobian determinant:

$${}^1j = \det {}^1\mathbf{J} = (x_8 - x_4)(y_5 - y_4) - (x_5 - x_4)(y_8 - y_4)$$

■ Jacobian inverse matrix:

$${}^1\mathbf{J}^{-1} = \frac{1}{{}^1j} \begin{bmatrix} x_8 - x_4 & -x_5 + x_4 \\ -y_8 + y_4 & y_5 - y_4 \end{bmatrix}$$

Concrete example: bilinear triangular element (2d)

■ Local mass matrix:

$$\begin{aligned} {}^1\mathbf{M} &= \int_{\Omega^a} \rho {}^a\mathbf{H}^T {}^a\mathbf{H} {}^1j \, d\xi \\ &= \int_{\Omega^a} \rho \begin{bmatrix} {}^ah_1^2\mathbf{I} & {}^ah_1{}^ah_2\mathbf{I} & {}^ah_1{}^ah_3\mathbf{I} \\ {}^ah_2{}^ah_1\mathbf{I} & {}^ah_2^2\mathbf{I} & {}^ah_2{}^ah_3\mathbf{I} \\ {}^ah_3{}^ah_1\mathbf{I} & {}^ah_3{}^ah_2\mathbf{I} & {}^ah_3^2\mathbf{I} \end{bmatrix} {}^1j \, d\xi_2 d\xi_1 \end{aligned}$$

■ Local stiffness matrix:

$${}^1\mathbf{K} = \int_{\Omega^a} \begin{bmatrix} \cdots & ({}^1\mathbf{J}^{-1}\nabla_{\xi} {}^ah_i)^T \mathbf{C} ({}^1\mathbf{J}^{-1}\nabla_{\xi} {}^ah_j) & \cdots \\ \cdots & & \cdots \end{bmatrix} {}^1j \, d\xi_2 d\xi_1$$

Assembly of the mass matrix: local to global

$${}^1\mathbf{M} = \int_{\Omega^a} \rho \begin{bmatrix} {}^a h_1^2 \mathbf{I} & {}^a h_1 {}^a h_2 \mathbf{I} & {}^a h_1 {}^a h_3 \mathbf{I} \\ {}^a h_2 {}^a h_1 \mathbf{I} & {}^a h_2^2 \mathbf{I} & {}^a h_2 {}^a h_3 \mathbf{I} \\ {}^a h_3 {}^a h_1 \mathbf{I} & {}^a h_3 {}^a h_2 \mathbf{I} & {}^a h_3^2 \mathbf{I} \end{bmatrix} {}^e j d\xi_2 d\xi_1 = \begin{bmatrix} {}^1\mathbf{M}_{11} & {}^1\mathbf{M}_{12} & {}^1\mathbf{M}_{13} \\ {}^1\mathbf{M}_{21} & {}^1\mathbf{M}_{22} & {}^1\mathbf{M}_{23} \\ {}^1\mathbf{M}_{31} & {}^1\mathbf{M}_{32} & {}^1\mathbf{M}_{33} \end{bmatrix}$$

We assembly local mass matrices into the global ones using the assembly operator:

$$\mathbf{M} = \bigcup_{e=1}^m {}^e\mathbf{M} = \sum_{e=1}^m {}^e\mathbf{L}^T {}^e\mathbf{M} {}^e\mathbf{L}$$

where

$${}^1\mathbf{L} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

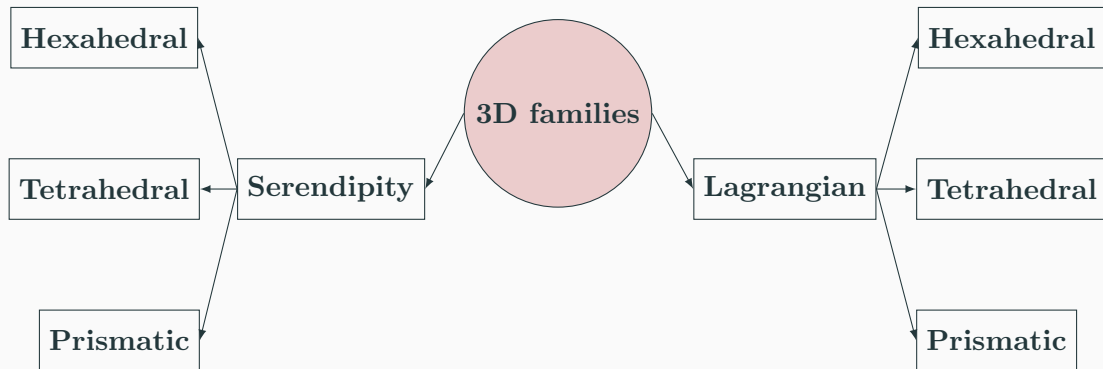
Node id. local	Node id. global
1	4
2	8
3	5

Assembly of the mass matrix: local to global

$${}^1\mathbf{L}^T {}^1\mathbf{M} {}^1\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & {}^1\mathbf{M}_{11} & {}^1\mathbf{M}_{13} & 0 & 0 & {}^1\mathbf{M}_{12} \\ 0 & 0 & 0 & {}^1\mathbf{M}_{31} & {}^1\mathbf{M}_{33} & 0 & 0 & {}^1\mathbf{M}_{32} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & {}^1\mathbf{M}_{21} & {}^1\mathbf{M}_{23} & 0 & 0 & {}^1\mathbf{M}_{22} \end{bmatrix}$$

3D finite elements

Three-dimensional families of finite elements



Serendipity vs Lagrangian elements

Lagrangian finite elements

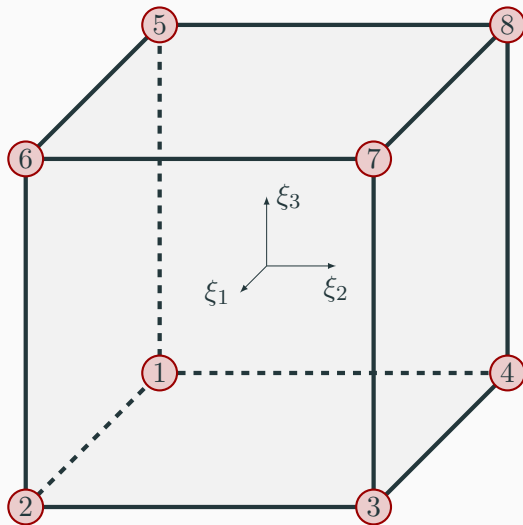
- Defined using Lagrange polynomials.
- Nodes are placed at both *element edges* and *inside the element*.
- Suitable for higher-order accuracy in both structured and unstructured meshes.

Serendipity finite elements

- Defined using a reduced number of nodes compared to Lagrangian elements.
- Nodes exist *only on the edges* (no interior nodes for 2D elements or nodes on faces for 3D elements).
- More computationally efficient but less accurate for high-order approximations.

Trilinear hexahedral Lagrangian element

Node	Coordinates
1	$(-1, -1, -1)$
2	$(+1, -1, -1)$
3	$(+1, +1, -1)$
4	$(-1, +1, -1)$
5	$(-1, -1, +1)$
6	$(+1, -1, +1)$
7	$(+1, +1, +1)$
8	$(-1, +1, +1)$



Base functions for trilinear Lagrangian hexahedral elements

$$^ah_1 = 0.125(1 - \xi_1)(1 - \xi_2)(1 - \xi_3)$$

$$^ah_2 = 0.125(1 + \xi_1)(1 - \xi_2)(1 - \xi_3)$$

$$^ah_3 = 0.125(1 + \xi_1)(1 + \xi_2)(1 - \xi_3)$$

$$^ah_4 = 0.125(1 - \xi_1)(1 + \xi_2)(1 - \xi_3)$$

$$^ah_5 = 0.125(1 - \xi_1)(1 - \xi_2)(1 + \xi_3)$$

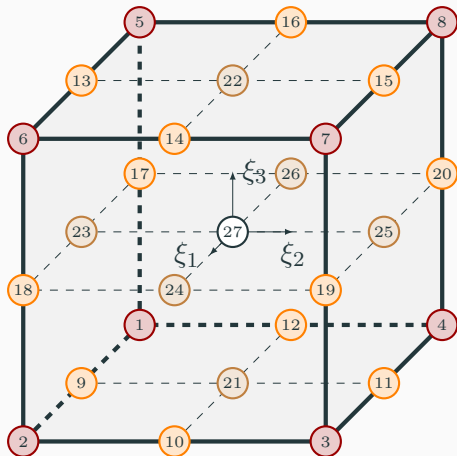
$$^ah_6 = 0.125(1 + \xi_1)(1 - \xi_2)(1 + \xi_3)$$

$$^ah_7 = 0.125(1 + \xi_1)(1 + \xi_2)(1 + \xi_3)$$

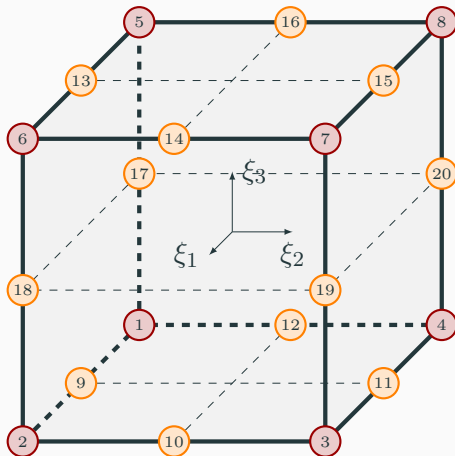
$$^ah_8 = 0.125(1 - \xi_1)(1 + \xi_2)(1 + \xi_3)$$

Triquadratic hexahedral elements

Lagrangian



Serendipity



Trilinear tetrahedral Lagrangian element

Node	Coordinates
1	(0, 0, 0)
2	(1, 0, 0)
3	(0, 1, 0)
4	(0, 0, 1)

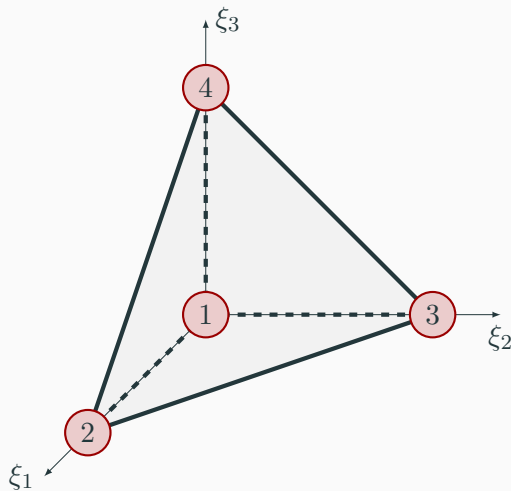
Base functions

$$^a h_1 = 1 - \xi_1 - \xi_2 - \xi_3$$

$$^a h_2 = \xi_1$$

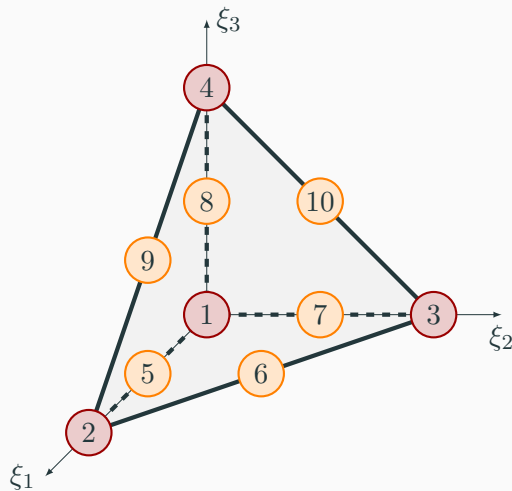
$$^a h_3 = \xi_2$$

$$^a h_4 = \xi_3$$



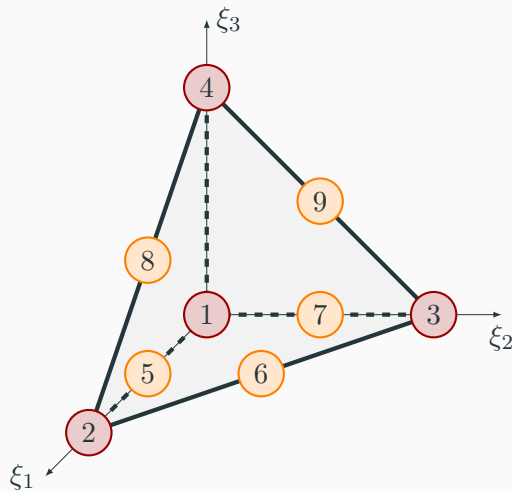
Triquadratic tetrahedral Lagrangian element

Node	Coordinates
1	(0, 0, 0)
2	(1, 0, 0)
3	(0, 1, 0)
4	(0, 0, 1)
5	(0.5, 0, 0)
6	(0.5, 0.5, 0)
7	(0, 0.5, 0)
8	(0, 0, 0.5)
9	(0.5, 0, 0.5)
10	(0, 0.5, 0.5)



Triquadratic tetrahedral serendipity element

Node	Coordinates
1	(0, 0, 0)
2	(1, 0, 0)
3	(0, 1, 0)
4	(0, 0, 1)
5	(0.5, 0, 0)
6	(0.5, 0.5, 0)
7	(0, 0.5, 0)
8	(0.5, 0, 0.5)
9	(0, 0.5, 0.5)



Trilinear prismatic Lagrangian element

Node	Coordinates
1	$(0, 0, -1)$
2	$(1, 0, -1)$
3	$(0, 1, -1)$
4	$(0, 0, 1)$
5	$(1, 0, 1)$
6	$(0, 1, 1)$

Base functions

$$^a h_1 = (1 - \xi_1 - \xi_2)(1 - \xi_3)/2$$

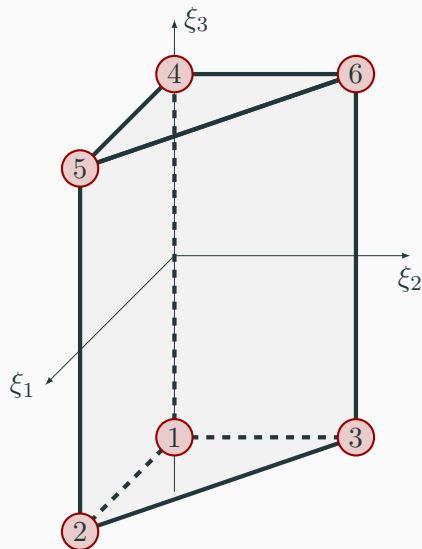
$$^a h_2 = \xi_1(1 - \xi_3)/2$$

$$^a h_3 = \xi_2(1 - \xi_3)/2$$

$$^a h_4 = (1 - \xi_1 - \xi_2)(1 + \xi_3)/2$$

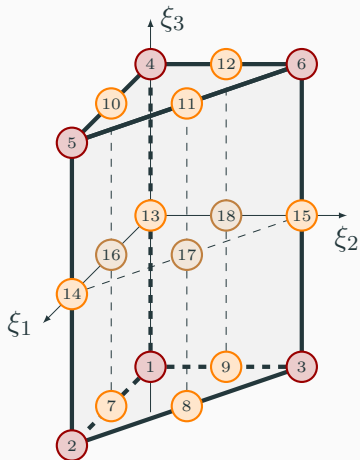
$$^a h_5 = \xi_1(1 + \xi_3)/2$$

$$^a h_6 = \xi_2(1 + \xi_3)/2$$

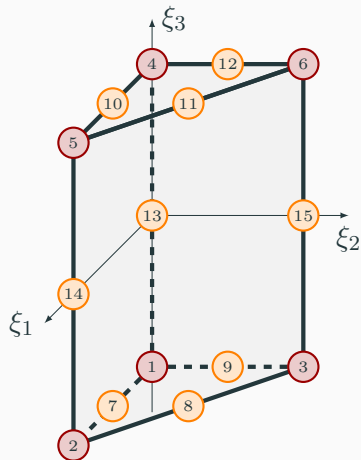


Triquadratic prismatic elements

Lagrangian



Serendipity



Numerical integration

Gauss-Legendre quadrature

- Numerical integration helps automate the calculation of the elementary quantities ${}^e\mathbf{K}$, ${}^e\mathbf{M}$ and ${}^e\mathbf{r}$.
- Integration is approximated as:

$$\int_{\Omega^a} f(\boldsymbol{\xi}) d\boldsymbol{\xi} \approx \sum_{i=1}^{r_1} \sum_{j=1}^{r_2} \sum_{k=1}^{r_3} \omega_i^1 \omega_j^2 \omega_k^3 f(\xi_1^i, \xi_2^j, \xi_3^k)$$

where ξ_j^i are known as *Gauss points* and ω_i^j are their associated *Gauss weights*.

👉 If f is a polynomial of degree $d_i = 2r_i - 1$ in the variable ξ_i , then the Gauss-Legendre approximation formula is **exact**.

Numerical integration over hexahedral solid elements

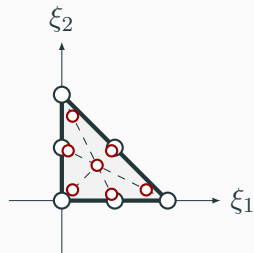
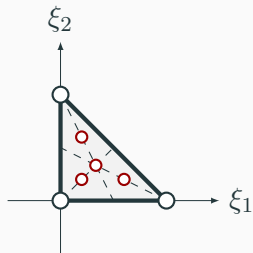
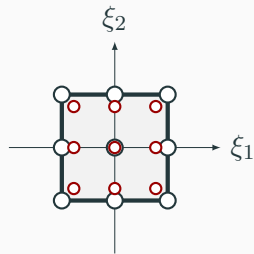
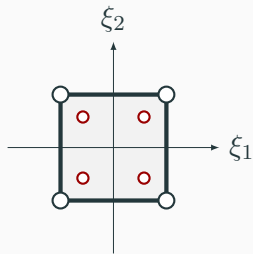
This allows us to compute the stiffness matrix, the mass matrix and the load vector numerically as:

$${}^e\mathbf{K} \approx \sum_{i=1}^{r_1} \sum_{j=1}^{r_2} \sum_{k=1}^{r_3} \omega_i^1 \omega_j^2 \omega_k^3 \left[({}^e\mathbf{J}^{-1} \nabla_{\xi} {}^a\mathbf{H})^T \mathbf{C} ({}^e\mathbf{J}^{-1} \nabla_{\xi} {}^a\mathbf{H})^e j \right]_{\xi_1=\xi_1^i, \xi_2=\xi_2^j, \xi_3=\xi_3^k}$$

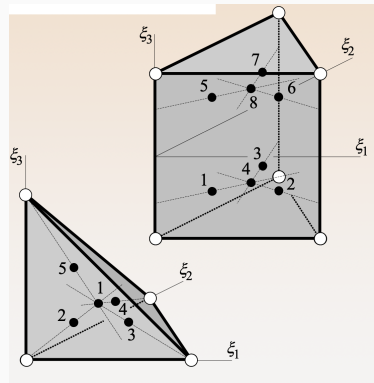
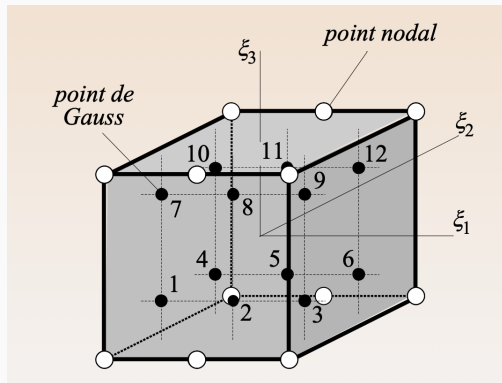
$${}^e\mathbf{M} \approx \sum_{i=1}^{r_1} \sum_{j=1}^{r_2} \sum_{k=1}^{r_3} \omega_i^1 \omega_j^2 \omega_k^3 \left[\rho {}^a\mathbf{H}^T {}^a\mathbf{H} \right]_{\xi_1=\xi_1^i, \xi_2=\xi_2^j, \xi_3=\xi_3^k}$$

$${}^e\mathbf{r} \approx \sum_{i=1}^{r_1} \sum_{j=1}^{r_2} \tilde{\omega}_i^1 \tilde{\omega}_j^2 \left[{}^a\mathbf{H}^T \hat{\mathbf{f}} \right]_{\xi_1=\tilde{\xi}_1^i, \xi_2=\tilde{\xi}_2^j} + \sum_{i=1}^{r_1} \sum_{j=1}^{r_2} \sum_{k=1}^{r_3} \omega_i^1 \omega_j^2 \omega_k^3 \left[{}^a\mathbf{H}^T \mathbf{f} \right]_{\xi_1=\xi_1^i, \xi_2=\xi_2^j, \xi_3=\xi_3^k}$$

An illustration of Gauss points - 2d elements

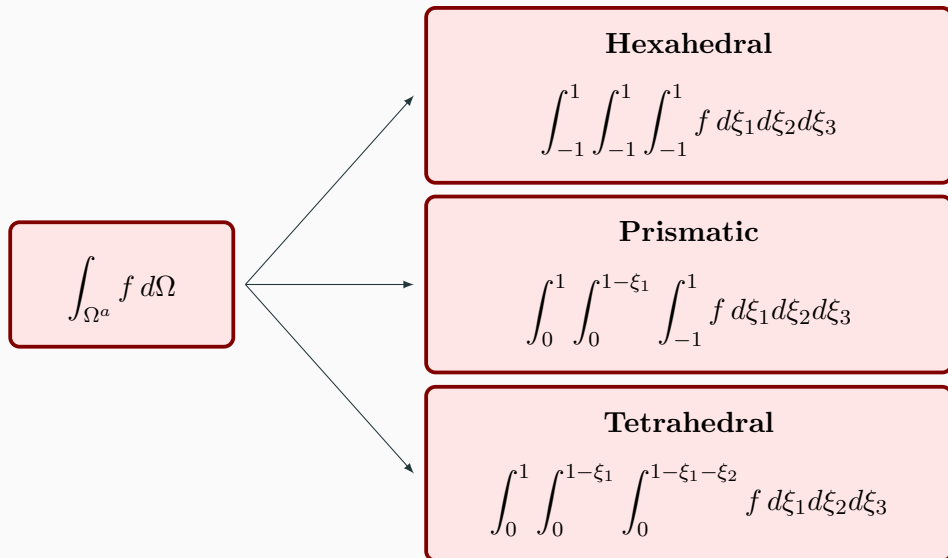


An illustration of Gauss points - 3d elements



(Credit: Thomas Gmür - Dynamique numérique des solides et des structures)

Numerical integration in 3d



Gauss-Legendre numerical integration

Hexahedral

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 f d\xi_1 d\xi_2 d\xi_3$$



$$\approx \sum_{i=1}^{r_1} \sum_{j=1}^{r_2} \sum_{k=1}^{r_3} \omega_i^1 \omega_j^2 \omega_k^3 f(\xi_1^i, \xi_2^j, \xi_3^k)$$

Prismatic

$$\int_0^1 \int_0^{1-\xi_1} \int_{-1}^1 f d\xi_1 d\xi_2 d\xi_3$$



$$\approx \sum_{i=1}^{r_1} \sum_{k=1}^{r_2} \omega_i^{12} \omega_k^3 f(\xi_1^i, \xi_2^i, \xi_3^k)$$

Tetrahedral

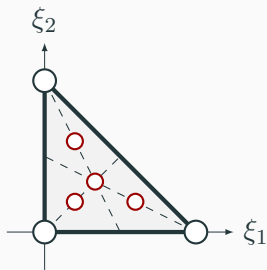
$$\int_0^1 \int_0^{1-\xi_1} \int_0^{1-\xi_1-\xi_2} f d\xi_1 d\xi_2 d\xi_3$$



$$\approx \sum_{i=1}^{r_1} \omega_i^{123} f(\xi_1^i, \xi_2^i, \xi_3^i)$$

Example of numerical integration using Gauss-Legendre quadrature in 2d

$$\int_{\Omega^a} f(\xi_1, \xi_2) d\xi_1 d\xi_2 \approx \sum_{i=1}^4 \omega_i f(\xi_1^i, \xi_2^i)$$



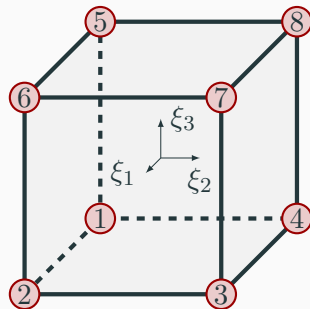
Gauss points (ξ_1^i, ξ_2^i)	Weights (ω_i)
$(1/5, 1/5)$	$25/96$
$(3/5, 1/5)$	$25/96$
$(1/5, 3/5)$	$25/96$
$(1/3, 1/3)$	$9/32$

Since we are using 2 Gauss points in each *direction*, the approximation is exact for polynomials of degree up to 3 in each variable.

Number of Gauss points for exact integration

Trilinear hexahedral element

- Basis functions are linear in each variable.
- Exact integration with:
 - $2 \times 2 \times 2$ Gauss points for ${}^e\mathbf{M}$
 - $1 \times 1 \times 1$ Gauss points for ${}^e\mathbf{K}$if the element is not distorted.
- Note that in case of distortion:
 - Jacobian is a function of $\boldsymbol{\xi}$: ${}^e j = {}^e j(\boldsymbol{\xi})$
 - ${}^e \mathbf{J}^{-1} = {}^e \mathbf{J}^{-1}(\boldsymbol{\xi})$ function of $\boldsymbol{\xi}$ at the denominator.
 - $\frac{\partial {}^e \mathbf{H}}{\partial \mathbf{x}} = {}^e \mathbf{J}^{-1} \frac{\partial {}^e \mathbf{H}}{\partial \boldsymbol{\xi}}$ is not a polynomial function of $\boldsymbol{\xi}$ anymore.



👉 Gauss quadrature will never be exact in case of distortion.

Number of Gauss points for exact integration

Rule for exact integration: evaluate the number of Gauss points on the archetypal element Ω^a (not on ${}^e\Omega$). Let n_i be the maximum number of nodes in direction ξ_i .

- **Hexahedral:** $n_1 \times n_2 \times n_3$ Gauss points.
- **Prismatic:** $n_{12} \times n_3$ Gauss points.
- **Tetrahedral:** n_{123} Gauss points

Example of exact integration for **mass matrix**:

- Trilinear Hexahedral: $n_1 \times n_2 \times n_3 = 2 \times 2 \times 2$
- Trilinear Tetrahedral: $n_{123} = 5$
- Trilinear Prismatic: $n_{12} \times n_3 = 4 \times 2$

Comparison of exact and reduced integration

Exact integration

- ✓ Integrals are computed exactly
- ✓ Monotonicity of convergence
- ✗ Overestimation of stiffness
- ✗ Risk of locking in linear elements

Reduced integration

- ✓ Reduction of computational costs
- ✗ Loss of monotonicity of convergence
- ✗ Hourglass effect in hexahedral structured meshes

Selective reduced integration:

- involves using fewer integration points in certain areas and more in others, based on the specific requirements of the analysis,
- reduce computational costs while maintaining accuracy in critical areas,
- mitigate the hourglass effect.

MATLAB PDE Toolbox

Examples

- ① Error analysis: exact vs numerical integration
- ② Tuning fork example from MATLAB PDE Toolbox
- ③ Transient linear analysis of cantilever beam

► [Go to Matlab Drive](#)

Complementary topics

Triquadratic hexahedral serendipity base functions

1. Vertex nodes (1-8):

$${}^a h_i = 0.125 \cdot (1 \pm \xi_1) \cdot (1 \pm \xi_2) \cdot (1 \pm \xi_3)$$

2. Mid-edge nodes (9-20):

$${}^a h_i = \begin{cases} 0.25 \cdot (1 - \xi_1^2) \cdot (1 \pm \xi_2) \cdot (1 \pm \xi_3) \\ 0.25 \cdot (1 \pm \xi_1) \cdot (1 - \xi_2^2) \cdot (1 \pm \xi_3) \\ 0.25 \cdot (1 \pm \xi_1) \cdot (1 \pm \xi_2) \cdot (1 - \xi_3^2) \end{cases}$$

3. Correction formula for vertex nodes (1-8):

$${}^a h_i \leftarrow {}^a h_i - 0.5 \cdot ({}^a h_{j_1} + {}^a h_{j_2} + {}^a h_{j_3})$$

→ Triquadratic hexahedral serendipity base functions

4. Mid-face nodes (21-26):

$${}^a h_i = \begin{cases} 0.5 \cdot (1 \pm \xi_1) \cdot (1 - \xi_2^2) \cdot (1 - \xi_3^2) \\ 0.5 \cdot (1 - \xi_1^2) \cdot (1 \pm \xi_2) \cdot (1 - \xi_3^2) \\ 0.5 \cdot (1 - \xi_1^2) \cdot (1 - \xi_2^2) \cdot (1 \pm \xi_3) \end{cases}$$

5. Vertex nodes (1-8) adjustment:

$${}^a h_i \leftarrow {}^a h_i - 0.5 \cdot ({}^a h_{j_1} + {}^a h_{j_2} + {}^a h_{j_3})$$

6. Mid-edge nodes (9-20) adjustment:

$${}^a h_i \leftarrow {}^a h_i - 0.5({}^a h_{j_1} + {}^a h_{j_2})$$

Triquadratic hexahedral base functions

7. Interior node (27):

$${}^a h_{27} = (1 - \xi_1^2)(1 - \xi_2^2)(1 - \xi_3^2)$$

8. Vertex nodes (1-8) adjustment:

$${}^a h_i \leftarrow {}^a h_i - 0.125 {}^a h_{27}$$

9. Mid-edge nodes (9-20) adjustment:

$${}^a h_i \leftarrow {}^a h_i + 0.25 {}^a h_{27}$$

10. Mid-face nodes (21-26) adjustment:

$${}^a h_i \leftarrow {}^a h_i - 0.5 {}^a h_{27}$$

→ **Triquadratic hexahedral base functions**

Base functions for tetrahedral elements

